Exposure and Demographic Context

Official Thermal Extremes and Population Aging in Hong Kong, 2013–2023

A Laidlaw Scholars deliverable preceding Hospital Authority outcome analysis

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This note reports only what can be defended with public data already in hand. It does not estimate associations between temperature and cardiovascular admissions. Those analyses await an approved Hospital Authority extract.

Positioning statement

As Laidlaw Scholars we are accountable for two things at once: intellectual ambition and evidentiary honesty. The wider project asks how officially defined thermal extremes—especially hot nights and cold days—relate to cardiovascular hospital burden in an aging Hong Kong. That question cannot be answered without clinical outcomes. What *can* be answered now, rigorously, is whether the environmental and demographic setting of 2013–2023 contains the contrast the study design requires, and whether that setting supports a coexistence framing rather than a "heat replaced cold" narrative.

This deliverable therefore ships three products: (i) validated official extreme-day series at the Hong Kong Observatory (HKO) Headquarters; (ii) a spell-structure decomposition of those extremes; and (iii) Census and Statistics Department (C&SD) mid-year population change in intervention-relevant older age bands. All three are **Real data**. Hospital Authority (HA) outcomes remain **Pending**.

1 Scientific framing

Earlier Hong Kong daily studies found cold-dominant cardiovascular associations goggins2013,goggins2012. Recent work shows that official binary hot nights may be a weak biological metric relative to nighttime heat intensity guo2024, while consecutive extreme events carry additional mortality information wang2019. Our planned monthly design estimates hospital *burden*, not daily triggering; coefficients will not be numerically comparable to Goggins-era per-°C rate ratios.

Against that literature, the present note asks a narrower, prior question:

> *Do contemporary Hong Kong thermal extremes and age structure provide meaningful exposure and demographic contrast for a 2013–2023 monthly study that keeps both heat and cold in view?*

A negative answer would argue for redesign. A positive answer does not prove a heat effect; it justifies proceeding to outcome analysis without overclaiming.

2 Data and methods

2.1 Thermal extremes

Daily HKO Headquarters observations were parsed from official `dailyExtract` files and aggregated to calendar months. Official definitions were applied without modification:

Metric	Definition
Hot night	Daily minimum temperature ≥ 28°C
Very hot day	Daily maximum temperature ≥ 33°C
Extremely hot day	Daily maximum temperature ≥ 35°C
Cold day	Daily minimum temperature ≤ 12°C

Months with temperature completeness below 90% would have extreme counts set to missing; no such months occurred in the core window. Annual totals for hot nights, very hot days, and cold days were compared with HKO *The Year's Weather* summaries for 2013–2023: **33 of 33 annual checks matched**.

Spell metrics follow the spirit of Wang et al. wang2019: days belonging to runs of five or more consecutive hot nights (or very hot days), and months touched by a simple 2D3N-type window (at least two very hot days and three hot nights within five days). These are descriptive exposure engineering, not causal claims.

2.2 Population

Mid-year population by sex and five-year age group was imported from C&SD Table 110-01001 (MDT public file; values in thousands converted to persons). Ages 65–69 and 70–74 are kept separate, following collaborator advice that these cohorts are both growing and practically reachable for intervention. Aging contrasts use exact mid-year counts for 2013 and 2023.

2.3 What this note deliberately excludes

- Any Hospital Authority admission counts or model coefficients.
- Placeholder air-pollution series (awaiting real EPD imports and Hogan quality review).
- Claims that heat has replaced cold, or that climate change has already shifted the cardiovascular risk regime.

3 Results

3.1 Thermal extremes: rising heat, persistent cold

Figure 1 shows annual counts of hot nights, very hot days, and cold days. Hot nights rose from 10 in 2013 to 61 in 2021 (56 in 2023). Very hot days rose from 17 to 54 over the same 2013–2021 comparison.

Cold days did not disappear: 14 in 2013, 13 in 2021, and 14 in 2023.

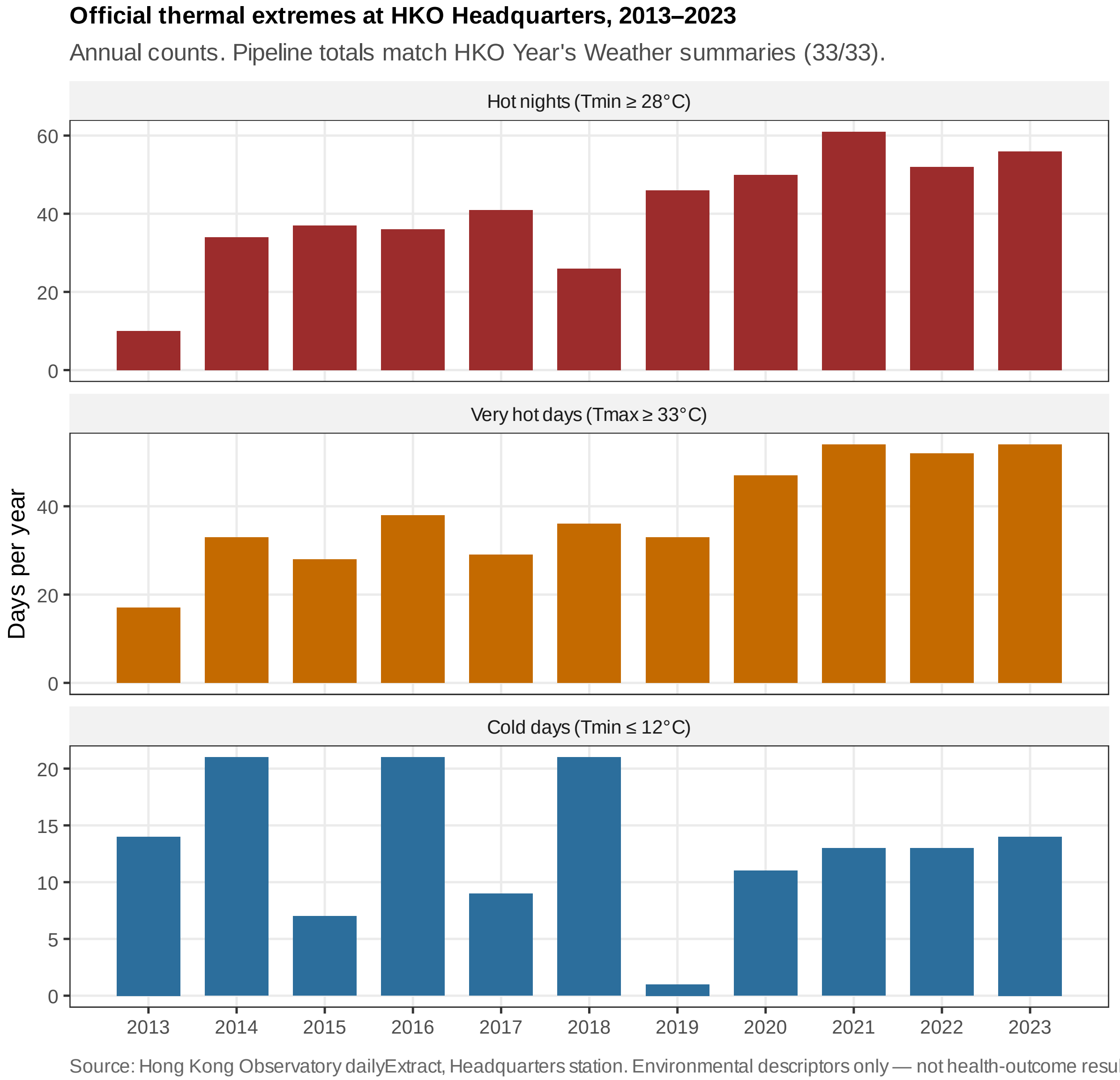

Figure 1: Annual official thermal extremes at HKO Headquarters, 2013–2023. Pipeline annual totals match HKO *Year's Weather* summaries for all 33 checks. These are environmental exposure descriptors, not health-outcome results.

Period means sharpen the contrast without implying a formal trend test (Table 1). Mean annual hot nights increased from 30.7 (2013–2018) to 53.0 (2019–2023). Mean cold days declined from 15.5 to 10.4 but remained non-trivial. The scientific implication is coexistence with changing weights—not succession.

Figure 2 shows the same coexistence at monthly resolution. Hot-night bars intensify in the later period; cold-day lines continue to appear in winter months of warm years. For a monthly burden model, this pattern supplies exposure variation on both margins.

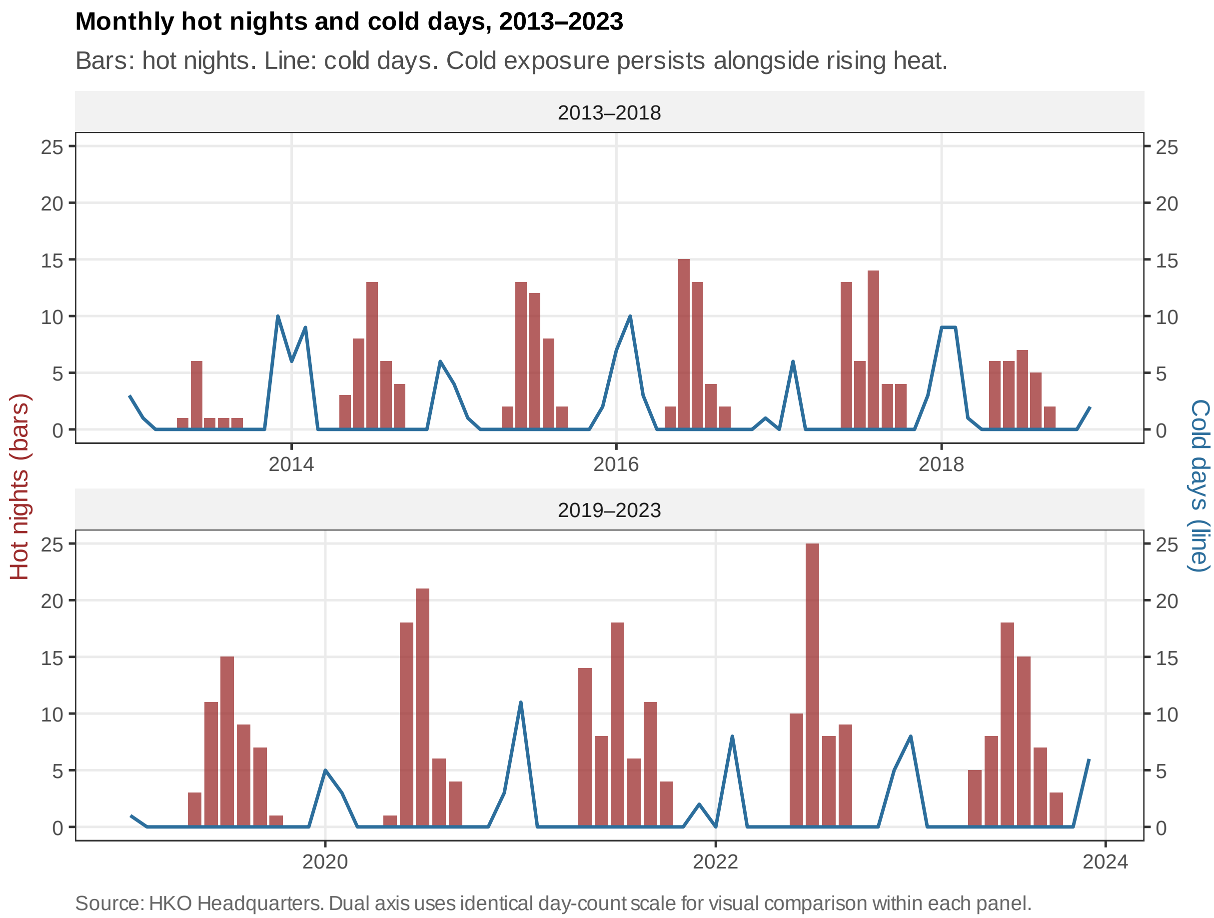

Figure 2: Monthly hot nights (bars) and cold days (line), 2013–2023, split by period. Cold days remain visible alongside the later rise in hot nights.

Table 1: Period means of annual extreme counts, HKO Headquarters

Period	Years	Mean hot nights	Mean very hot days	Mean cold days
2013–2018	6	30.7	30.2	15.5
2019–2023	5	53.0	48.0	10.4

3.2 Spell structure: extremes are not only scattered counts

Monthly totals discard sequence. Across 2013–2023, on average about 42% of annual hot nights fell inside spells of five or more consecutive nights (Figure 3). That share was higher in 2019–2023 (mean 0.51) than in 2013–2018 (mean 0.34). Months touched by a 2D3N-type window also rose (means 2.7 vs 4.4 months per year).

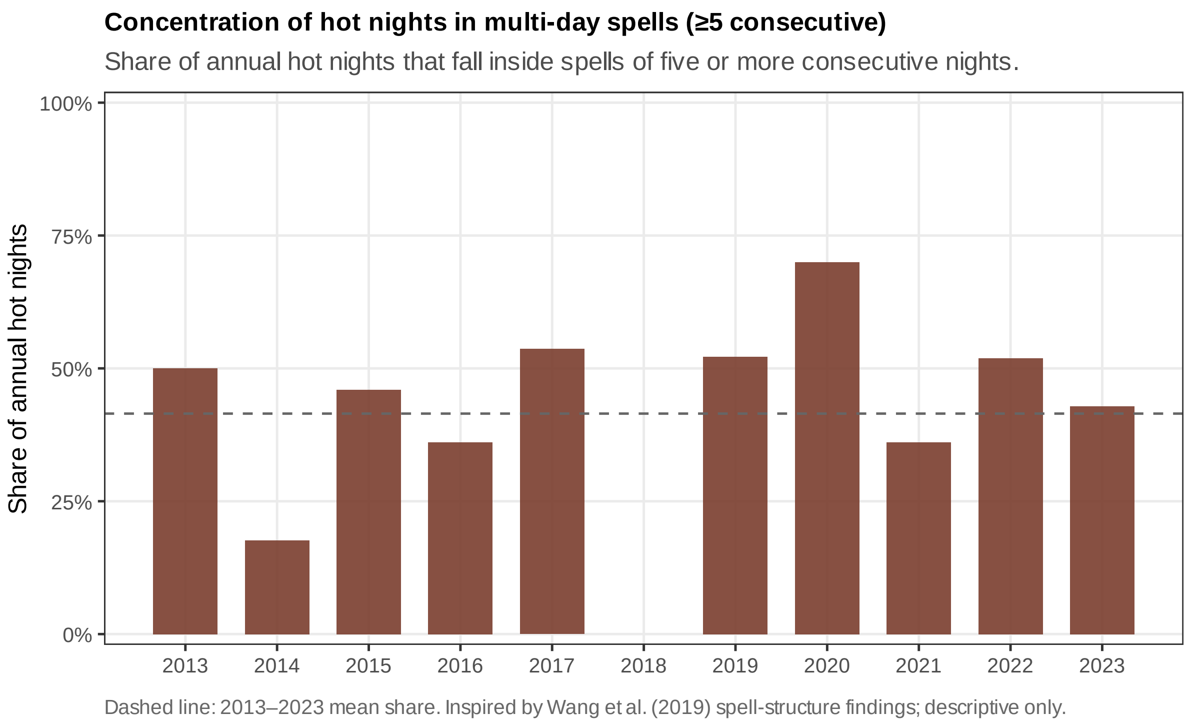

Figure 3: Share of annual hot nights occurring inside consecutive spells of length ≥ 5. Dashed line: 2013–2023 mean. Descriptive exposure structure only.

We read this cautiously. Spell concentration motivates retaining spell and combination metrics as *prespecified alternatives* to official binary counts—especially given Guo et al.'s finding that binary HNday28 may not track hospitalization risk while intensity does guo2024. A null monthly association for official hot-night counts would be compatible with that literature and would not close the nighttime-heat question.

3.3 Population aging: 65–69 and 70–74 nearly doubled

C&SD mid-year counts show rapid growth in the older bands prioritized by the team (Table 2; Figures 4–5). Ages 65–69 rose from 295,100 in mid-2013 to 568,500 in mid-2023 (+92.6%). Ages 70–74 rose from 213,100 to 424,800 (+99.3%). Ages 85+ grew by 63.9%. Several mid-life bands were flat or declining.

For the eventual admission analysis, these denominators are not decorative. Raw counts of AMI or

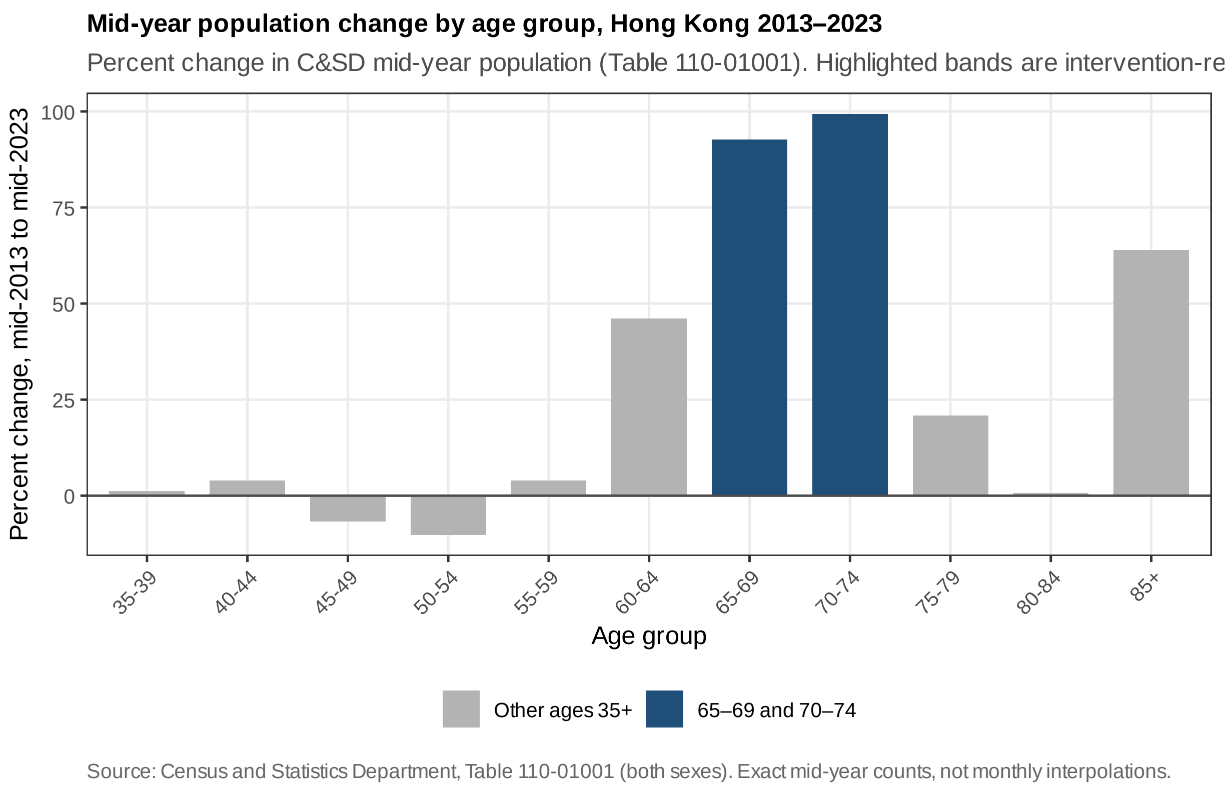

Figure 4: Percent change in mid-year population by age group, 2013–2023. Ages 65–69 and 70–74 are highlighted as intervention-relevant cohorts.

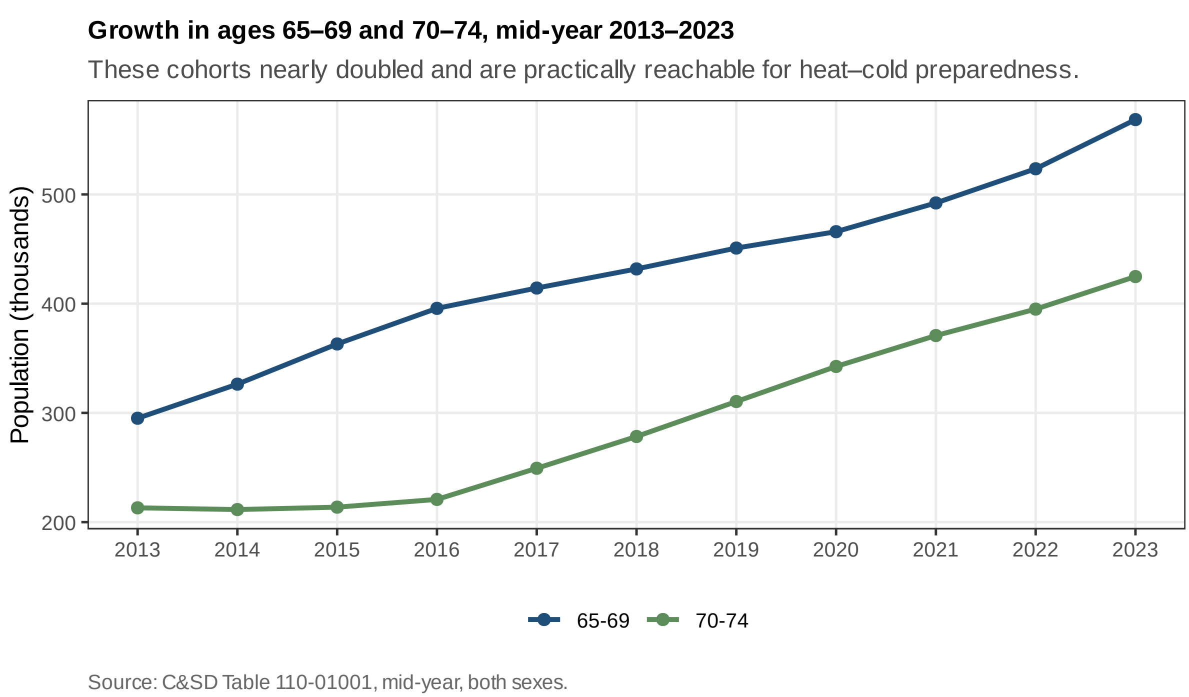

Figure 5: Mid-year population trajectories for ages 65–69 and 70–74, 2013–2023.

Table 2: Mid-year population in older age groups, C&SD Table 110-01001 (both sexes)

Age group	Mid-2013	Mid-2023	Absolute change	% change
65–69	295,100	568,500	+273,400	+92.6
70–74	213,100	424,800	+211,700	+99.3
75–79	210,200	254,100	+43,900	+20.9
80–84	157,600	158,600	+1,000	+0.6
85+	143,900	235,900	+92,000	+63.9

stroke admissions cannot be interpreted as changing risk while the older population nearly doubles. The planned offset—log(age–sex population $\times$ days in month)—is therefore a scientific requirement, not a modelling flourish.

4 Reflective synthesis

Three conclusions follow from public data alone.

1. The exposure window is analytically usable. 2013–2023 is not a uniformly hot decade. It contains substantial heat contrast and retained cold days. A monthly model has something to estimate on both margins.

2. Coexistence is the honest framing. Cold days in high-heat years (for example 13 cold days alongside 61 hot nights in 2021) make a "regime shift already completed" story premature. Our working question remains whether an emerging heat burden coexists with persistent cold-related risk in an aging city.

3. Demography amplifies the stakes of whatever associations exist. Even a stable per-person cold association would imply a larger absolute hospital burden as ages 65–69 and 70–74 expand. Heat associations, if present, would land on a larger vulnerable base. Separating those bands in the HA extract is therefore both scientifically and operationally justified.

We also record what we do *not* yet know. We do not know whether official hot-night counts will associate with principal-diagnosis AMI, ischemic stroke, or hemorrhagic stroke at monthly scale. We do not know whether spell metrics will outperform binary counts. We do not know how COVID-era care-seeking will reshape apparent associations. Those are empirical questions for the HA stage—not gaps to be filled by speculation.

5 Limitations

- Headquarters temperature is a territory-wide proxy; microclimate and housing modify personal exposure.
- Monthly aggregation aligns with official extreme counts and demographic offsets but attenuates short-lag triggering, especially for heat.
- Spell and 2D3N indicators are simplified constructions from daily minima/maxima; they are not hourly hot-night excess (HNe).

- Population figures include foreign domestic helpers in Table 110-01001; a sensitivity using the excluding-FDH table remains available if HA residency definitions require it.
- No pollution adjustment is attempted here because real EPD series are not yet finalized.

6 Implications for the next stage

1. **Roro / HA.** Proceed with the principal-diagnosis, diagnosis-stratified extract; keep 65–69 and 70–74 separate; confirm coding, episodes, and suppression rules before regression.
2. **Hogan / EPD.** Import real general-station NO_2, O_3, $PM_{2.5}$, and PM_{10}; retain staged models and cautious ozone handling.
3. **Analysis plan.** Freeze the pre-analysis plan now: co-primary hot nights and cold days; spell/intensity alternatives; null-interpretation rule compatible with Guo et al. guo2024; no direct numeric comparison with Goggins per-°C estimates.
4. **Manuscript.** This note can seed the Exposure Results section. Outcome Results and a definitive Discussion await HA data.

Closing

Laidlaw research is judged not only by how far a question reaches, but by how carefully a team refuses to outrun its evidence. With HKO and C&SD data we can say—defensibly—that Hong Kong's 2013–2023 thermal and demographic setting supports a coexistence study of official extremes and cardiovascular hospital burden in an aging city. We cannot yet say what that burden's weather associations are. The next mile is outcome validity, not additional rhetoric.

Reproducibility

Figures and tables were produced by `scripts/16_exposure_aging_deliverable.R` from `data_processed/climate_mo` and `data_raw/csd_population/csd_population_age_sex_annual_normalized.csv`. HKO annual validation: 33/33 match to *Year's Weather*. Pipeline mode: `run_pipeline_dev.R` (synthetic HA rehearsal only; not used in this note's empirical claims).

References

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